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(54) **TUNED DAMPING METHOD FOR SEISMIC ISOLATION OF PILE FOUNDATIONS IN DEEP-WATER LONG-SPAN CONTINUOUS RIGID FRAME BRIDGE**

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(57) **ABSTRACT**

Some embodiments of the disclosure disclose a tuned damping method for seismic isolation of pile foundations in a deep-water long-span continuous rigid frame bridge, which relates to the technical field of shock absorption for bridge engineering structures. It solves the problem that the "pile foundation seismic isolation" of deep-water long-span continuous rigid frame bridges lacks a theoretical quantitative calculation method, making it impossible to fully exert the effect of "pile foundation isolation". The present disclosure includes: simplifying a deep-water long-span continuous rigid frame bridge model into a 2-degree-of-freedom dynamical model; setting an optimization objective function based on the 2-degree-of-freedom dynamical model; performing parameter optimization based on the optimization objective function; and determining structural seismic parameters of the deep-water long-span continuous rigid frame bridge based on parameters optimized in step 3.

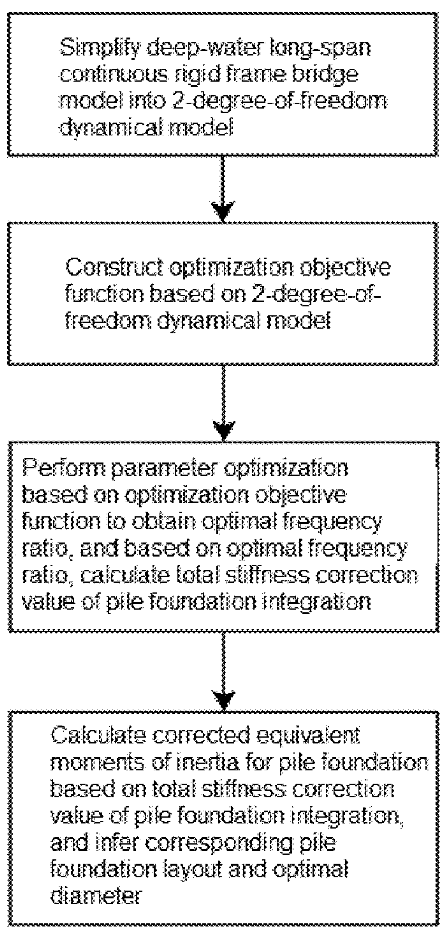


FIG. 1

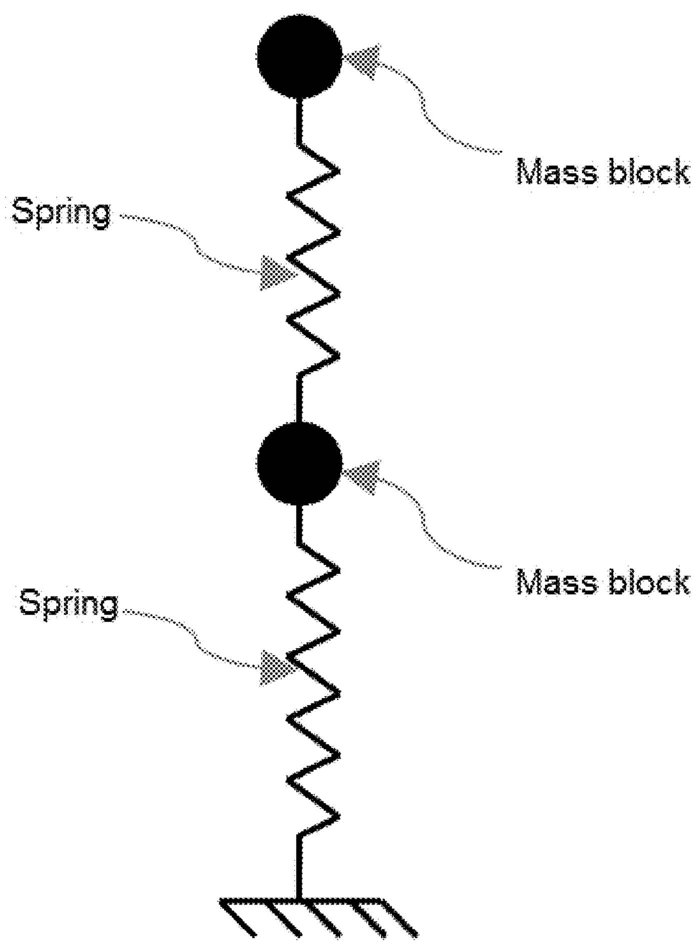


FIG. 2

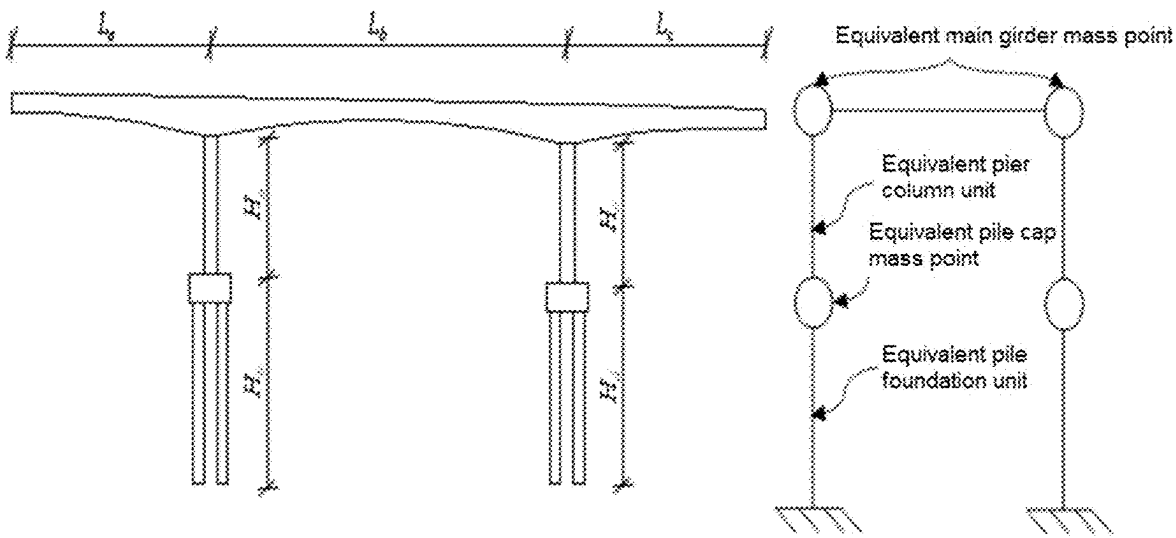


FIG. 3(a)

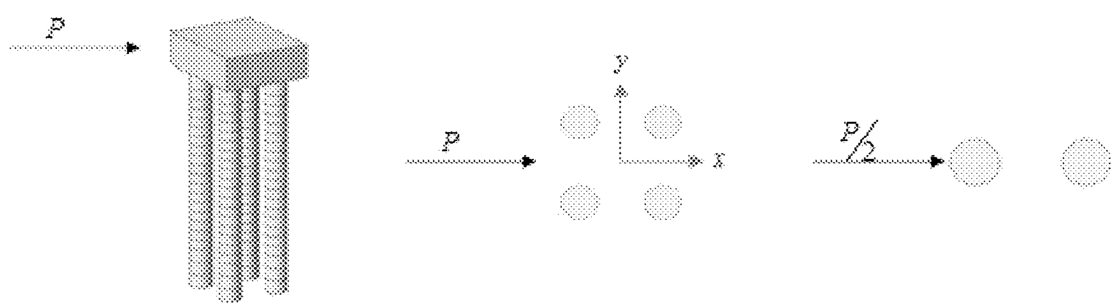


FIG. 3(b)

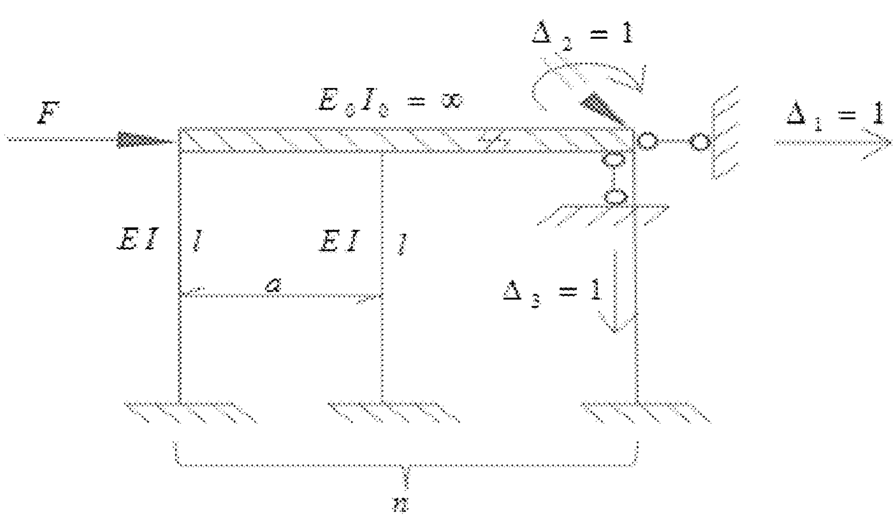


FIG. 4

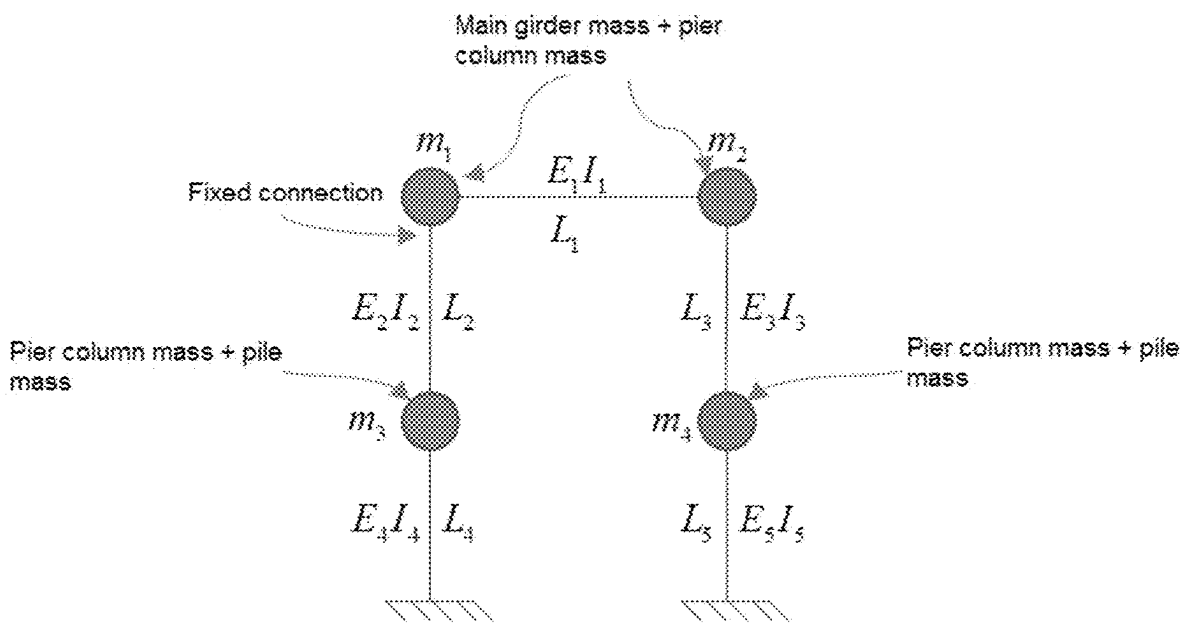

Main girder mass + pier column mass
m_1
m_2
Fixed connection
E_1I_1
L_1
E_2I_2 L_2
L_3 E_3I_3
Pier column mass + pile mass
Pier column mass + pile mass
m_3
m_4
E_4I_4 L_4
L_5 E_5I_5

FIG. 5

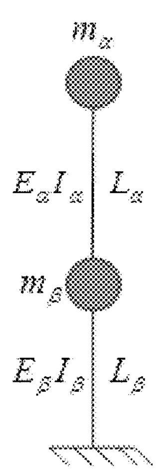

m_α
$E_\alpha I_\alpha$ L_α
m_β
$E_\beta I_\beta$ L_β

FIG. 6

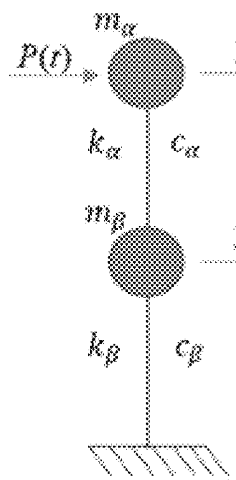

m_α
$P(t)$
y_1
k_α
c_α
m_β
y_2
k_β
c_β

FIG. 7

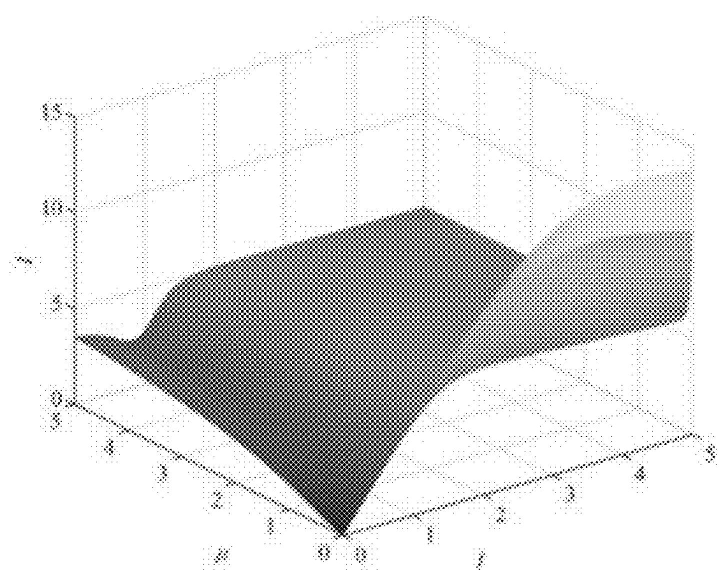

15
10
5
0
5
4
3
2
1
0
0
1
2
3
4
5

FIG. 8

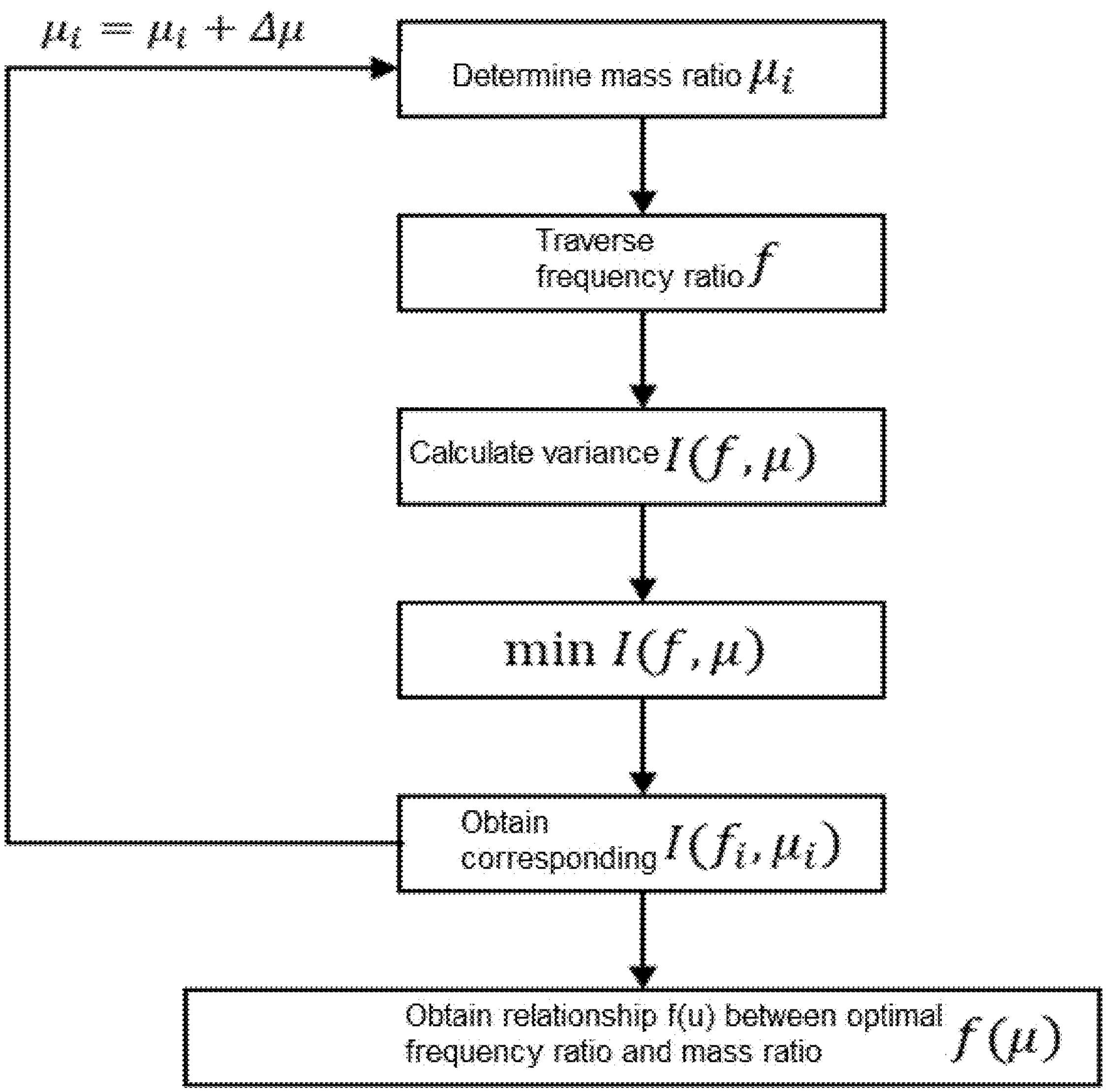

$\mu_i = \mu_i + \Delta\mu$
Determine mass ratio μ_i
Traverse frequency ratio f
Calculate variance $I(f, \mu)$
$\min I(f, \mu)$
Obtain corresponding $I(f_i, \mu_i)$
Obtain relationship f(u) between optimal frequency ratio and mass ratio $f(\mu)$

FIG. 9

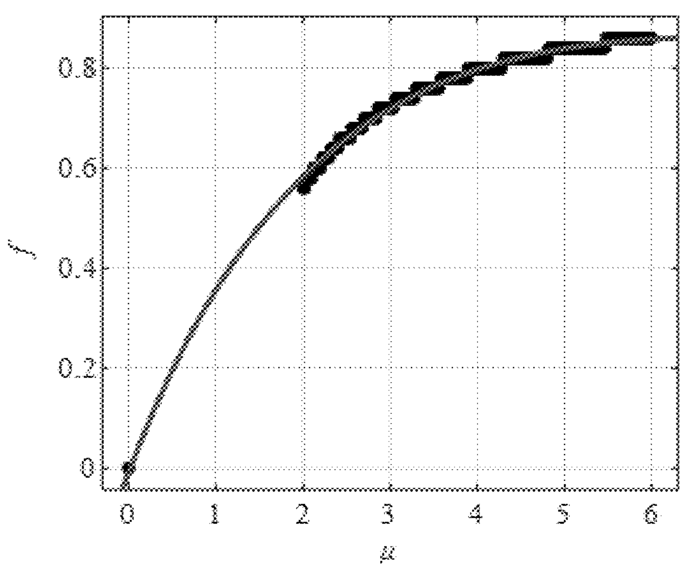
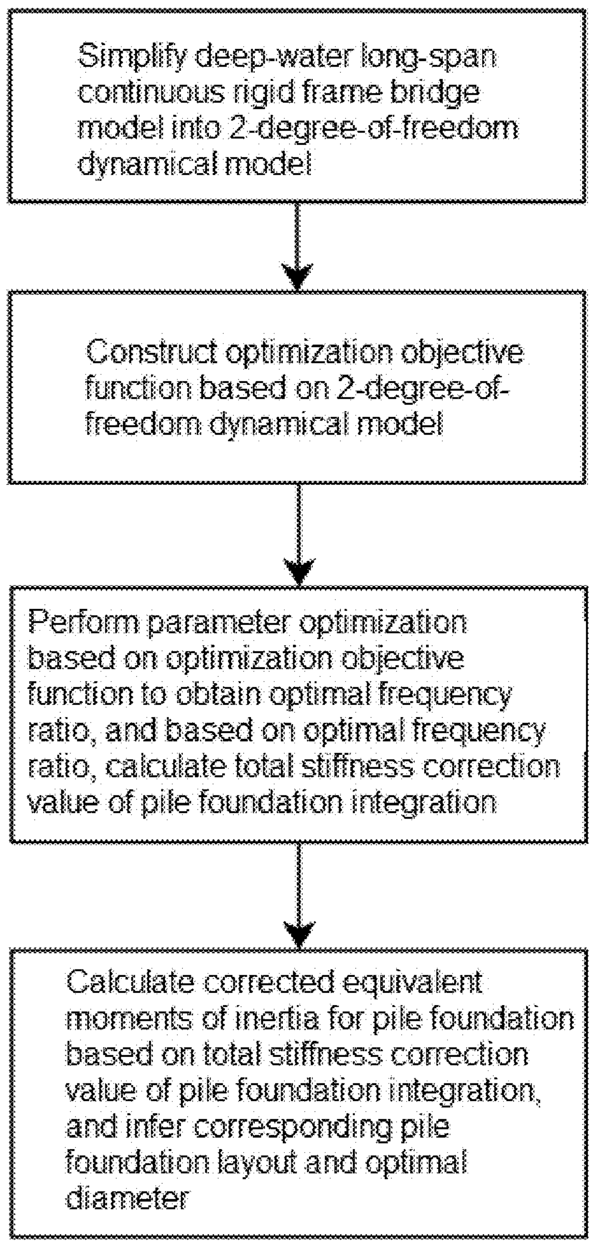


FIG. 10


Simplify deep-water long-span continuous rigid frame bridge model into 2-degree-of-freedom dynamical model
Construct optimization objective function based on 2-degree-of-freedom dynamical model
Perform parameter optimization based on optimization objective function to obtain optimal frequency ratio, and based on optimal frequency ratio, calculate total stiffness correction value of pile foundation integration
Calculate corrected equivalent moments of inertia for pile foundation based on total stiffness correction value of pile foundation integration, and infer corresponding pile foundation layout and optimal diameter

TUNED DAMPING METHOD FOR SEISMIC ISOLATION OF PILE FOUNDATIONS IN DEEP-WATER LONG-SPAN CONTINUOUS RIGID FRAME BRIDGE

TECHNICAL FILED

[0001] The present disclosure relates to the field of seismic damping technology for bridge engineering structures, and specifically relates to a tuned damping method for seismic isolation of pile foundations in a deep-water long-span continuous rigid frame bridge.

BACKGROUND

[0002] Seismic isolation design achieves the goal of reducing the internal force response of structures by pre-installing seismic isolation elements and prolonging the structural period. It is an important technical approach in the seismic design of bridges. In conventional seismic isolation design, bearings are commonly used as seismic isolation elements. In deep-water long-span continuous rigid frame bridges, it is inconvenient to use seismic isolation bearings to achieve seismic isolation due to the fixed connection between a main pier and a main girder. However, the deep-water long-span continuous rigid frame bridges typically use high-pile caps with relatively long unrestrained pile lengths and relatively weak lateral stiffness of pile foundations, which can prolong the natural vibration period of the structure, thereby forming "pile foundation seismic isolation".

[0003] The engineering community has recognized the seismic isolation effect of pile foundations. Through the proper matching of superstructure mass-bridge pier stiffness-pile cap mass-pile foundation stiffness, more seismic energy can be converted into structural kinetic energy, further enhancing the "pile foundation seismic isolation" effect. However, the discussion on this effect remains at a qualitative level. There remains a lack of theoretical research and quantitative calculation methods to effectively utilize the beneficial effect, optimize the structural design, and fully exploit the seismic isolation effect of pile foundations.

SUMMARY

[0004] To solve the problems in the art known to inventors mentioned above, the present disclosure provides a tuned damping method for seismic isolation of pile foundations in a deep-water long-span continuous rigid frame bridge to solve the problem of the lack of theoretical quantitative calculation methods for "pile foundation seismic isolation" in deep-water long-span continuous rigid frame bridges.

[0005] A tuned damping method for seismic isolation of pile foundations in a deep-water long-span continuous rigid frame bridge, including the following steps:

[0006] step 1: simplifying a deep-water long-span continuous rigid frame bridge model into a 2-degree-of-freedom dynamical model;

[0007] step 2: constructing an optimization objective function based on the 2-degree-of-freedom dynamical model;

[0008] step 3: performing parameter optimization based on the optimization objective function to obtain an optimal frequency ratio f, and based on the optimal frequency ratio f to calculate corrected pile foundation stiffness $k_{\beta 1}$;

[0009] step 4: calculating corrected equivalent moments of inertia I_{41} and I_{51} for a pile foundation based on the corrected pile foundation stiffness $k_{\beta 1}$, and inferring a corresponding pile foundation layout and an optimal diameter based on the corrected equivalent moments of inertia I_{41} and I_{51}.

[0010] In some embodiments, the step 1 includes:

[0011] step 1.1: simplifying the deep-water long-span continuous rigid frame bridge model into a 3-degree-of-freedom dynamical model;

[0012] step 1.2: simplifying the 3-degree-of-freedom dynamical model into the 2-degree-of-freedom dynamical model.

[0013] In some embodiments, the step 1.1 includes:

[0014] letting anti-thrust stiffness of a pile group in the deep-water long-span continuous rigid frame bridge model be k_p, and solving by means of a displacement method or finite element simulation method to obtain an equivalent moment of inertia I_p:

$$I_p = \frac{k_p L_p^3}{12 E_p} \tag{1}$$

[0015] where, L_p is a pile foundation length, and E_p is the elastic modulus of pile foundation;

[0016] basing on the equivalent moment of inertia I_p to simplify the deep-water long-span continuous rigid frame bridge model into the 3-degree-of-freedom dynamical model, with parameters comprising the pier column stiffness, the pile foundation stiffness, and the mass parameters, and specific formulas for the parameters are as follows:

[0017] The pier column stiffness comprises a pier column longitudinal stiffness k_2 and a pier column transverse stiffness k_3:

$$k_2 = \frac{3E_2 I_2}{L_2^3} + \Delta k \tag{2}$$

$$k_3 = \frac{3E_3 I_3}{L_3^3} + \Delta k$$

[0018] The pile foundation stiffness comprises a pile foundation longitudinal stiffness k_4 and a pile foundation transverse stiffness k_5, which can be calculated using a force method or a finite element method;

Main Girder Bending Restraint Correction Term:

$$\Delta k = \frac{6E_2 I_2 (6E_1 I_1 L_2 + E_2 I_2 L_1)}{L_2^3 (3E_1 I_1 L_2 + 2E_2 I_2 L_1)} - \frac{3E_2 I_2}{L^3} \tag{3}$$

[0019] the mass parameters comprises a pier top mass and a pier bottom mass:

The Pier Top Mass:

$$m_1 = \frac{m_b}{2} + \frac{m_{p2}}{2} \quad (4)$$
$$m_2 = \frac{m_b}{2} + \frac{m_{p3}}{2}$$

The Pier Bottom Mass:

$$m_3 = \frac{m_c}{2} + \frac{m_{p2}}{2} + \frac{m_{p4}}{2} \quad (5)$$
$$m_4 = \frac{m_c}{2} + \frac{m_{p3}}{2} + \frac{m_{p5}}{2}$$

[0020] where, I_2 and I_3 are moments of inertia of cross-sections of pier columns, I_4 and I_5 are calculated according to equation (1), m_1 is the sum of main girder mass and pier column mass on a left span of a bridge, m_2 is the sum of main girder mass and pier column mass on a right span of the bridge, m_3 is the sum of pier column mass and pile foundation mass on the left span of the bridge, m_4 is the sum of pier column mass and pile foundation mass on the right span of the bridge, m_b is the total mass of a main girder, m_{p2} and m_{p3} are masses of bridge piers respectively, m_{p4} and m_{p5} are masses of pile foundations respectively, L_1 is the length of the main girder, L_2 and L_3 are the lengths of the bridge piers, L_4 and L_5 are the lengths of the pile foundations, E_1 is the elastic modulus of the main girder, E_2 and E_3 are the elastic modulus of the bridge piers, and L_4 and L_5 are the elastic modulus of the pile foundations;

[0021] a mass matrix M and a stiffness matrix K of the 3-degree-of-freedom dynamical model are:

$$M = \begin{bmatrix} m_1 + m_2 & 0 & 0 \\ 0 & m_3 & 0 \\ 0 & 0 & m_4 \end{bmatrix} \quad (6)$$

$$K = \begin{bmatrix} k_2 + k_3 + 2\Delta k & -k_2 - \Delta k & -k_3 - \Delta k \\ -k_2 - \Delta k & k_4 + k_2 + \Delta k & 0 \\ -k_3 - \Delta k & 0 & k_5 + k_3 + \Delta k \end{bmatrix} \quad (7)$$

[0022] In some embodiments, the step 1.2 includes: assuming the deep-water long-span continuous rigid frame bridge model to be a symmetric structure, that is: $E_2I_2=E_3I_3$, $E_4I_4=E_5I_5$, $m_1=m_2$, $m_3=m_4$, $L_2=L_3$, and $L_4=L_5$, and further simplifying the 3-degree-of-freedom dynamical model into the 2-degree-of-freedom dynamical model, with the mass matrix M and stiffness matrix K thereof as shown in equations (8) and (9) respectively:

$$M = \begin{bmatrix} M_\alpha & 0 \\ 0 & M_\beta \end{bmatrix} \quad (8)$$

-continued

$$K = \begin{bmatrix} k_\alpha & -k_\alpha \\ -k_\alpha & k_\alpha + k_\beta \end{bmatrix} \quad (9)$$

[0023] where,

$$L_\alpha = \frac{L_2 + L_3}{2},\ L_\beta = \frac{L_4 + L_5}{2} \quad (10)$$

$$M_\alpha = m_1 + m_2,\ M_\beta = m_3 + m_4 \quad (11)$$

$$k_\alpha = 4(k_2 + k_3) \quad (12)$$

$$k_\beta = k_4 + k_5 \quad (13)$$

[0024] where, k_α is the total stiffness of a 2-degree-of-freedom pier column integration, and k_β is the total stiffness of the 2-degree-of-freedom pile foundation integration.

[0025] In some embodiments, the step 2 includes:

[0026] step 2.1: constructing motion equations for a 2-degree-of-freedom system based on the 2-degree-of-freedom dynamical model;

[0027] Step 2.2: constructing an optimization objective function based on the motion equations for the 2-degree-of-freedom system.

[0028] In some embodiments, the step 2.1 includes: considering damping of the bridge piers and the pile foundations, and enabling the bridge structure to undergo random vibration under the action of P(t), wherein y_1 and y_2 represent 2-degree-of-freedom displacement time history functions relative to a foundation, and motion equations for the 2-degree-of-freedom dynamical model can be obtained by means of a direct equilibrium method as follows:

$$m_\alpha \ddot{y}_\alpha + c_\alpha(\dot{y}_\alpha - \dot{y}_\beta) + k_\alpha(y_\alpha - y_\beta) = P(t) \quad (14)$$
$$m_\beta \ddot{y}_\beta + c_\beta \dot{y}_\beta + c_\alpha(\dot{y}_\beta - \dot{y}_\alpha) + k_\beta y_\beta + k_\alpha(y_\beta - y_\alpha) = 0$$

[0029] where c_i is the damping coefficient; y_i is the relative displacement of the mass point; $\dot{u}_i$ is the relative velocity of the mass point; $\ddot{y}_i$ is the relative acceleration of the mass point; P(t) is an inertial force caused by ground motion, and P(t) is a random excitation which can be decomposed into the superposition of a series of harmonic components as follows:

$$P(t) = \int_{-\infty}^{\infty} p_0 e^{i\theta t} d\theta \quad (15)$$

[0030] for any frequency component θ, $P^{(\theta)}(t)=p_0e^{i\theta t}$ is a harmonic excitation, and resulting displacements $y_1^{(6)}$ (t) and $y_2^{(\theta)}(t)$ are also harmonic variables and can be expressed as follows:

$$y_1^{(\theta)}(t) = Y_1 e^{i\theta t},\ y_2^{(\theta)}(t) = Y_2 e^{i\theta t} \quad (16)$$

[0031] substituting $P^{(\Theta)}(t)$ and equation (16) into equation (14) to obtain

$$-m_\alpha\theta^2 Y_1 + c_\alpha i\theta(Y_1 - Y_2) + k_\alpha(Y_1 - Y_2) = p_0 \tag{17}$$
$$-m_\beta\theta^2 Y_2 + c_\beta i\theta Y_2 + c_\alpha i\theta(Y_2 - Y_1) + k_\beta Y_\beta + k_\alpha(Y_2 - Y_1) = 0$$

[0032] solving to obtain amplitudes of $y_1(t)$ and $y_2(t)$:

$$Y_1 = \frac{-p_0(k_\alpha + k_\beta - m_\beta\theta^2 + c_\alpha\theta i + c_\beta\theta i)}{\begin{gathered}k_\alpha m_\alpha\theta^2 - k_\alpha k_\beta + k_\alpha m_\beta\theta^2 + k_\beta m_\alpha\theta^2 + c_\alpha c_\beta\theta^2 - m_\alpha m_\beta\theta^4 - \\ k_\beta c_\alpha\theta i - k_\alpha c_\beta\theta i + m_\alpha c_\alpha\theta^3 i + m_\beta c_\alpha\theta^3 i + m_\alpha c_\beta\theta^3 i\end{gathered}} \tag{18}$$

$$Y_2 = \frac{-p_0(k_\alpha + c_\alpha\theta i)}{\begin{gathered}k_\alpha m_\alpha\theta^2 - k_\alpha k_\beta + k_\alpha m_\beta\theta^2 + k_\beta m_\alpha\theta^2 + c_\alpha c_\beta\theta^2 - m_\alpha m_\beta\theta^4 - \\ k_\beta c_\alpha\theta i - k_\alpha c_\beta\theta i + m_\alpha c_\alpha\theta^3 i + m_\beta c_\alpha\theta^3 i + m_\alpha c_\beta\theta^3 i\end{gathered}} \tag{19}$$

[0033] substituting into equations (16) to obtain $y_1^{(\Theta)}(t)$ and $y_2^{(\Theta)}(t)$, and summing $y_1^{(\Theta)}(t)$ and $y_2^{(\Theta)}(t)$ over a frequency domain to obtain $y_1(t)$ and $y_2(t)$.

[0034] In some embodiments, the step 2.2 includes: letting a pier bottom bending moment as an indicator to evaluate an overall structural response:

$$M(t) = k_\alpha L_\alpha(y_1(t) - y_2(t)) \tag{20}$$

[0035] for any frequency component θ,

$$M^{(\theta)}(t) = k_\alpha L_\alpha(y_1^{(\theta)}(t) - y_2^{(\theta)}(t)) \tag{21}$$

[0036] substituting equations (18) and (19) into equation (21) to obtain $M^{(\Theta)}(t)$, and expressing $M^{(\Theta)}(t)$ in the form of a transfer function H(θ) as follows:

$$M(t) = \int_{-\infty}^{\infty} H(\theta)P_0 e^{i\theta t}d\theta \tag{22}$$

where,

$$H(\theta) = k_\alpha L_\alpha \frac{k_\beta - m_\beta\theta^2 + ic_\beta\theta}{\begin{gathered}-k_\alpha m_\alpha\theta^2 + k_\alpha k_\beta - k_\alpha m_\beta\theta^2 - \\ k_\beta m_\alpha\theta^2 - c_\alpha c_\beta\theta^2 + m_\alpha m_\beta\theta^4 + \\ i(k_\beta c_\alpha\theta + k_\alpha c_\beta\theta - m_\alpha c_\alpha\theta^3 - m_\alpha c_\beta\theta^3)\end{gathered}} \tag{23}$$

[0037] when a variance σ_m^2 of M(t) is minimized, M(t) is also minimized, and let the variance σ_m^2 be:

$$\sigma_m^2 = E[|M(t)|^2] \tag{24}$$

[0038] substituting equation (23) into equation (24) and performing non-dimensionalization to obtain a non-dimensional form I of the variance σ_m^2:

$$I = \frac{1}{2\pi}\int_{-\infty}^{\infty}|H(g)|^2 dg \tag{25}$$

[0039] where, g represents a ratio

$$\left(\frac{\theta}{\omega_\alpha}\right)$$

of a seismic excitation to a structural frequency; and H(g) is the non-dimensional form of the transfer function H(θ):

$$H(g) = \frac{(f^2 - g^2) + 2i\frac{c_\beta}{c_{c\beta}}gf}{\begin{gathered}\left(-\frac{1}{\mu}g^2 + (g^2 - f^2)(g^2 - 1) + \frac{4c_\alpha c_\beta}{c_{c\alpha}c_{c\beta}}fg^2\right) + \\ i\left(2\frac{c_\alpha}{c_{c\alpha}}g\left(f^2 - \frac{1}{\mu}g^2 - g^2\right) + 2\frac{c_\beta}{c_{c\beta}}fg(1 - g^2)\right)\end{gathered}} \tag{26}$$

where,

$$\mu = \frac{m_\beta}{m_\alpha}\ \ \omega_\alpha^2 = \frac{k_\alpha}{m_\alpha}\ \ \omega_\beta^2 = \frac{k_\beta}{m_\beta}\ \ f = \frac{\omega_\beta}{\omega_\alpha} \tag{27}$$
$$g = \frac{\theta}{\omega_\alpha}c_{c\alpha} = 2m_\alpha\omega_\beta c_{c\beta} = 2m_\beta\omega_\beta\frac{k_\beta}{k_\alpha} = \mu f^2$$

[0040] In equation (26), μ represents a mass ratio of a pile cap to a main girder equivalent mass point, ω_α represents a natural vibration frequency of the main girder and the bridge pier, ω_β represents a natural vibration frequency of the pile cap and the pile foundation, f represents a frequency ratio between the pile foundation and the pier column, g represents a frequency ratio between an external load excitation and the pier column, $c_{c\alpha}$ represents critical damping of the bridge pier, and $c_{c\beta}$ represents critical damping of the pile foundation;

[0041] equation (25) is organized as:

$$H(g) = \frac{B_0 + igB_1 - g^2B_2}{A_0 + igA_1 - g^2A_2 - ig^3A_3 + g^4A_4} \tag{28}$$

where,

$$B_0 = f^2;\ B_1 = 2\frac{c_\beta}{c_{c\beta}}f;\ B_2 = 1;\ A_0 = f^2;\ A_1 = 2\frac{c_\alpha}{c_{c\alpha}}f^2 + 2\frac{c_\beta}{c_{c\beta}}f \tag{29}$$
$$A_2 = \frac{1}{\mu} + f^2 + 1 - \frac{4c_\alpha c_\beta}{c_{c\alpha}c_{c\beta}}f;\ A_3 = 2\frac{1}{\mu}\frac{c_\alpha}{c_{c\alpha}} + 2\frac{c_\alpha}{c_{c\alpha}} + 2\frac{c\beta}{c_{c\beta}}f;\ A_4 = 1$$

[0042] equation (28) is substituted into equation (25) and then integrated and let

$$\xi_1 = \frac{c_1}{c_{c\alpha}} \text{ and } \xi_2 = \frac{c_2}{c_c\beta},$$

the non-dimensional form I of the variance σ^2 can be obtained, which is the optimization objective function:

$$I = \frac{\begin{bmatrix} -A_0A_1A_4B_2^2 - A_0A_3A_4(B_1^4 - 2B_0B_2) + \\ A_4B_0^2(A_1A_4 - A_2A_3) \end{bmatrix}}{2A_0A_4(A_0A_3^2 + A_1^2A_4 - A_1A_2A_3)} = \frac{M(\mu, f, \xi_1, \xi_2)}{N(\mu, f, \xi_1, \xi_2)} \tag{30}$$

[0043] In some embodiments, the step 3 includes: optimizing the variance by means of an enumeration method, determining an optimal frequency ratio f corresponding to each mass ratio μ when the variance is minimized, and performing curve fitting to obtain an explicit expression for the optimal frequency ratio f:

$$f = 1.108e^{-0.02755\mu} - 1.116e^{-0.4368\mu} \tag{31}$$

[0044] based on the optimal frequency ratio f, calculating a total stiffness correction value $k_{\beta 1}$ of a 2-degree-of-freedom pile foundation integration:

$$k_{\beta 1} = \frac{m_\beta k_\alpha f^2}{m_\alpha} \tag{32}$$

[0045] In some embodiments, the optimization process using the enumeration method includes the following steps:

- **[0046]** (1) determining the mass ratio μ_i;
- **[0047]** (2) traversing the frequency ratio f;
- **[0048]** (3) calculating the variance I(f, μ);
- **[0049]** (4) minimizing the variance I;
- **[0050]** (5) obtaining the corresponding $I(f_i, \mu_i)$;
- **[0051]** (6) updating $\mu_i=\mu_i+\Delta\mu$; and simultaneously obtaining the relationship f(u) between the optimal frequency ratio and the mass ratio.

[0052] In some embodiments, the step 4 includes:

- **[0053]** calculating the mass ratio μ according to equation (27), substituting the mass ratio μ into equation (31) to obtain an optimal frequency ratio f of the pile foundation, according to equation (32) to obtain the total stiffness correction value $k_{\beta 1}$ for the 2-degree-of-freedom pile foundation integration, calculating corrected bridge pier and pile foundation stiffness k_{41} and k_{51}, back-calculating the equivalent moments of inertia I_{41} and I_{51} of the pile foundation according to equation (1), and based on the equivalent moments of inertia I_{41} and I_{51}, inferring the corresponding pile foundation layout and the optimal diameter, where calculation formulas for k_{41} and k_{51} are as follows:

$$k_{41} = \frac{k_{\beta 1}k_4}{k_\beta} \tag{33}$$

$$k_{51} = \frac{k_{\beta 1}k_5}{k_\beta}$$

[0054] The beneficial effects of the present disclosure are as follows:

[0055] Based on the fundamental principle of tuned damping, the present disclosure proposes a reasonable calculation method for the pile foundation-pile cap-pier column design parameters, thereby improving the damping efficiency of a pile foundation seismic isolation system. By adopting an optimized pile foundation diameter determined through random excitation optimization, the responses of the pier columns, pile foundations, and main girders are significantly reduced, effectively controlling the seismic response of the continuous rigid frame bridges. The determination method for design parameters of pile foundations and bridge piers is shifted from relying on experience to utilizing structural dynamics calculations, which results in more accurate parameters and faster parameter determination, significantly shortening the design time and lowering design costs.

BRIEF DESCRIPTION OF THE DRAWINGS

[0056] FIG. **1** illustrates a mechanical model of a tuned mass damper according to Embodiment 1.

[0057] FIG. **2** illustrates a high-pile cap continuous rigid frame bridge and an equivalent model thereof according to Embodiment 1.

[0058] FIG. **3** (*a*) illustrates a pile group foundation being simplified to a one-dimensional structure according to Embodiment 1.

[0059] FIG. **3** (*b*) illustrates a simplified planar model of the pile group foundation according to Embodiment 1.

[0060] FIG. **4** illustrates a 3-degree-of-freedom simplified model for a continuous rigid frame bridge according to Embodiment 1.

[0061] FIG. **5** illustrates a 2-degree-of-freedom simplified model for a continuous rigid frame bridge according to Embodiment 1.

[0062] FIG. **6** illustrates a 2-degree-of-freedom system model according to Embodiment 1.

[0063] FIG. **7** illustrates a I(f, μ) three-dimensional diagram according to Embodiment 1.

[0064] FIG. **8** illustrates a flowchart of calculating an optimal frequency ratio according to Embodiment 1.

[0065] FIG. **9** illustrates a graph of the optimal frequency ratio according to Embodiment 1.

[0066] FIG. **10** illustrates a flow diagram of a tuned damping method for seismic isolation of pile foundations in a deep-water long-span continuous rigid frame bridge.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0067] In order to make the objectives, technical solutions and advantages of embodiments of the present application clearer, the technical solutions in the embodiments of the present application will be described clearly and completely below with reference to the accompanying drawings in the embodiments of the present application. Apparently, the embodiments described are some of, rather than all of, the embodiments of the present application. Thus, the following detailed description of the embodiments of the present disclosure, as represented in the drawings, is not intended to limit the scope of the present application as claimed, but is merely representative of the selected embodiments of the present application. Based on the embodiments of the present application, all other embodiments obtained by those of ordinary skill in the art without creative work fall within the scope of protection of the present application.

Embodiment 1

[0068] A specific embodiment of the present disclosure will be described below in conjunction with FIGS. **1** to **10**.

[0069] A core component for mass-tuned damping is a tuned mass damper (TMD), and a mechanical model thereof is composed of mass blocks, a damper, and springs, as shown in FIG. **1**. When the TMD is added to an original system, the original system vibrates under dynamic loads, and the TMD generates an opposing force to counteract the vibration of the original system, thereby reducing the dynamic response.

[0070] For a deep-water long-span continuous rigid frame bridge, the mass is mainly distributed in a main girder and a pile cap. The mass of the pile cap can be regarded as the first mass M**1**, and the mass of the main girder can be regarded as the second mass M**2**. An unrestrained pile foundation in water can be regarded as a first spring K**1**, and a bridge pier can be regarded as a second spring K**2**. Therefore, a high-pile cap continuous rigid frame bridge can be equivalent to a TMD system, as shown in FIG. **2**. By applying the principles of mass-tuned damping, a reasonable stiffness ratio between the pile foundation and the bridge pier, as well as a reasonable mass ratio between the main girder and the pile cap, can be sought to reduce the seismic response of the structure. Based on the optimal stiffness or mass ratio, the seismic optimization design of the "pile foundation seismic isolation" system bridge can be guided.

[0071] A tuned damping method for seismic isolation of pile foundations in a deep-water long-span continuous rigid frame bridge specifically includes the following steps:

[0072] Step 1: simplifying a mechanical model;

[0073] step 1.1: simplifying a deep-water long-span continuous rigid frame bridge model into a 3-degree-of-freedom dynamical model;

[0074] In a direction perpendicular to seismic wave input, pile foundations are parallel, so the bending stiffness and axial stiffness thereof are the sum of the bending and axial stiffnesses of the pile foundations in the direction perpendicular to the seismic wave input. (as shown in FIG. **3**-*a*). Therefore, a pile foundation structure with a two-dimensional distribution can be simplified into a one-dimensional planar structure. Since the stiffness of a pile cap is much greater than the stiffness of a pile foundation, the pile cap can be considered a rigid body. Thus, a simplified planar model of a pile group foundation is shown in FIG. **3**-*b*.

[0075] Considering the axial deformation of the pile foundations, there are 3 key displacements. Even with n pile foundations along a seismic wave input direction, the number of basic unknown displacement quantities remains 3. Based on this, the anti-thrust stiffness k_p of a pile group and the equivalent moment of inertia I_p of the pile group can be solved:

$$I_p = \frac{k_p L_p^3}{12 E_p} \tag{1}$$

[0076] where, L_p and E_p represent the pile foundation length and the elastic modulus, respectively; k_p is the anti-thrust stiffness of the pile group; based on the simplified planar model of the pile group foundation, the simplified mechanical model for the continuous rigid frame bridge is shown in FIG. **4**.

[0077] Stiffness parameters in the figure include the pier column stiffness, the pile foundation stiffness, and the mass parameters, which are calculated according to formulas of structural dynamics methods as follows:

[0078] the pier column stiffness includes a pier column longitudinal stiffness and a pier column transverse stiffness:

$$k_2 = \frac{3E_2I_2}{L_2^3} + \Delta k \quad k_3 = \frac{3E_3I_3}{L_3^3} + \Delta k \tag{2}$$

[0079] the pile foundation stiffness comprises pile foundation longitudinal stiffness k_4 and pile foundation transverse stiffness k_5, which can be calculated using a force method or a finite element method;

Main Girder Bending Restraint Correction Term:

$$\Delta k = \frac{6E_2I_2(6E_1I_1L_2 + E_2I_2L_1)}{L_2^3(3E_1I_1L_2 + 2E_2I_2L_1)} - \frac{3E_2I_2}{L^3} \tag{3}$$

Pier Top Mass:

$$m_1 = \frac{m_b}{2} + \frac{m_{p2}}{2} \quad m_2 = \frac{m_b}{2} + \frac{m_{p3}}{2} \tag{4}$$

Pier Bottom Mass:

$$m_3 = \frac{m_c}{2} + \frac{m_{p2}}{2} + \frac{m_{p4}}{2} \tag{5}$$
$$m_4 = \frac{m_c}{2} + \frac{m_{p3}}{2} + \frac{m_{p5}}{2}$$

[0080] where, I_2 and I_3 are moments of inertia of cross-sections of pier columns, I_4 and I_5 are equivalent moments of inertia of the pile foundations and are calculated according to equation (1), m_1 is the sum of main girder mass and pier column mass on a left span of a bridge, m_2 is the sum of main girder mass and pier column mass on a right span of the bridge, m_3 is the sum of pier column mass and pile foundation mass on the left span of the bridge, m_4 is the sum of pier column mass and pile foundation mass on the right span of the bridge, m_b is the total mass of the main girder, m_{p2} and m_{p3} are masses of bridge piers, respectively, m_{p4} and m_{p5} are masses of pile foundations, L_1 is the length of the main girder, L_2 and L_3 are the lengths of the bridge piers, L_4 and L_5 are the lengths of the pile foundations, E_1 is the elastic modulus of the main girder, E_2 and E_3 are the elastic modulus of the bridge piers, and L_4 and L_5 are the elastic modulus of the pile foundations;

[0081] Since axial stiffness of the main girder L_1 is infinite, a 4-mass-point model shown in FIG. **4** is essentially a 3-degree-of-freedom dynamical model. The first degree of freedom corresponds to the horizontal displacement of m_1+m_2, the second degree of freedom corresponds to the horizontal displacement of m_3, and the third degree of freedom corresponds to the horizontal displacement of m_4. A mass matrix M and a stiffness matrix K of the model are:

$$M = \begin{bmatrix} m_1 + m_2 & 0 & 0 \\ 0 & m_3 & 0 \\ 0 & 0 & m_4 \end{bmatrix} \tag{6}$$

$$K = \begin{bmatrix} k_2 + k_3 + 2\Delta k & -k_2 - \Delta k & -k_3 - \Delta k \\ -k_2 - \Delta k & k_4 + k_2 + \Delta k & 0 \\ -k_3 - \Delta k & 0 & k_5 + k_3 + \Delta k \end{bmatrix} \tag{7}$$

[0082] step 1.2: simplifying the 3-degree-of-freedom dynamical model into a 2-degree-of-freedom dynamical model;

[0083] Clearly, in the 3-degree-of-freedom dynamical model shown in FIG. **4**, the motion of mass points m_3 and m_4 only occurs in two directions: in the same direction or the opposite directions. When moving in the same direction, an anti-symmetric vibration mode occurs, and when moving in the opposite directions, a symmetric vibration mode occurs. Since longitudinal seismic excitation is anti-symmetric, it only excites the anti-symmetric vibration mode. Therefore, under the longitudinal seismic action, the symmetric vibration mode does not contribute to the structural response. For simplicity in the derivation and from an engineering application perspective, it is assumed that the structure is symmetric, that is: $E_2I_2=E_3I_3$, $E_4I_4=E_5I_5$, $m_1=m_2$, $m_3=m_4$, $L_2=L_3$, and $L_4=L_5$, and the 3-degree-of-freedom dynamical model in FIG. **4** can be further simplified into a 2-degree-of-freedom dynamical model shown in FIG. **5**, with the mass matrix M and the stiffness matrix K thereof as shown in equations (8) and (9) respectively:

$$M = \begin{bmatrix} M_\alpha & 0 \\ 0 & M_\beta \end{bmatrix} \tag{8}$$

$$K = \begin{bmatrix} k_\alpha & -k_\alpha \\ -k_\alpha & k_\alpha + k_\beta \end{bmatrix} \tag{9}$$

where,

$$L_\alpha = \frac{L_2 + L_3}{2},\ L_\beta = \frac{L_4 + L_5}{2} \tag{10}$$

$$M_\alpha = m_1 + m_2,\quad M_\beta = m_3 + m_4 \tag{11}$$

$$k_\alpha = 4(k_2 + k_3) \tag{12}$$

$$k_\beta = k_4 + k_5 \tag{13}$$

[0084] where, k_α is the total stiffness of a 2-degree-of-freedom pier column integration, and k_β is total stiffness of the 2-degree-of-freedom pile foundation integration.

[0085] step 2: constructing an optimization objective function based on the 2-degree-of-freedom dynamical model;

[0086] step 2.1: constructing motion equations for a 2-degree-of-freedom system based on the 2-degree-of-freedom model;

[0087] the 2-degree-of-freedom system model is shown in FIG. **6**. Considering the damping of the bridge pier and the pile foundation, the structure undergoes random vibration under the action of P(t), where y_1 and y_2 represent the 2-degree-of-freedom displacement time history functions relative to the foundation.

[0088] Motion equations for the 2-degree-of-freedom dynamical model can be obtained by means of a direct equilibrium method:

$$\begin{aligned} & m_\alpha \ddot{y}_\alpha + c_\alpha(\dot{y}_\alpha - \dot{y}_\beta) + k_\alpha(y_\alpha - y_\beta) = P(t) \\ & m_\beta \ddot{y}_\beta + c_\beta \dot{y}_\beta + c_\alpha(\dot{y}_\beta - \dot{y}_\alpha) + k_\beta y_\beta + k_\alpha(y_\beta - y_\alpha) = 0 \end{aligned} \tag{14}$$

[0089] where c_i is the damping coefficient; y_i is the relative displacement of the mass point; $\dot{y}_i$ is the relative velocity of the mass point; $\ddot{y}_i$ is the relative acceleration of the mass point; P(t) is an inertial force caused by ground motion, and a random excitation, which can be decomposed into the superposition of a series of harmonic components as follows:

$$P(t) = \int_{-\infty}^{\infty} p_0 e^{i\theta t} d\theta \tag{15}$$

[0090] for any frequency component θ, $P^{(\Theta)}(t)=p_0e^{i\theta t}$ is a harmonic excitation, resulting displacements $y_1^{(\Theta)}(t)$ and $y_2^{(\Theta)}(t)$ are also harmonic variables and can be expressed as follows:

$$y_1^{(\theta)}(t) = Y_1 e^{i\theta t},\ y_2^{(\theta)}(t) = Y_2 e^{i\theta t} \tag{16}$$

[0091] Substituting $P^{(\Theta)}(t)$ and equation (17) into equation (15) to obtain

$$\begin{aligned} & -m_\alpha\theta^2 Y_1 + c_\alpha i\theta(Y_1 - Y_2) + k_\alpha(Y_1 - Y_2) = p_0 \\ & -m_\beta\theta^2 Y_2 + c_\beta i\theta Y_2 + c_\alpha i\theta(Y_2 - Y_1) + k_\beta Y_\beta + k_\alpha(Y_2 - Y_1) = 0 \end{aligned} \tag{17}$$

[0092] solving to obtain amplitudes of $y_1(t)$ and $y_2(t)$:

$$Y_1 = \frac{-p_0(k_\alpha + k_\beta - m_\beta\theta^2 + c_\alpha\theta i + c_\beta\theta i)}{\begin{array}{c} k_\alpha m_\alpha\theta^2 - k_\alpha k_\beta + k_\alpha m_\beta\theta^2 + k_\beta m_\alpha\theta^2 + c_\alpha c_\beta\theta^2 - m_\alpha m_\beta\theta^4 - \\ k_\beta c_\alpha\theta i - k_\alpha c_\beta\theta i + m_\alpha c_\alpha\theta^3 i + m_\beta c_\alpha\theta^3 i + m_\alpha c_\beta\theta^3 i \end{array}} \tag{18}$$

$$Y_2 = \frac{-p_0(k_\alpha + c_\alpha\theta i)}{\begin{array}{c} k_\alpha m_\alpha\theta^2 - k_\alpha k_\beta + k_\alpha m_\beta\theta^2 + k_\beta m_\alpha\theta^2 + c_\alpha c_\beta\theta^2 - m_\alpha m_\beta\theta^4 - \\ k_\beta c_\alpha\theta i - k_\alpha c_\beta\theta i + m_\alpha c_\alpha\theta^3 i + m_\beta c_\alpha\theta^3 i + m_\alpha c_\beta\theta^3 i \end{array}} \tag{19}$$

[0093] substituting into equations (16) to obtain $y_1^{(\Theta)}(t)$ and $y_2^{(\Theta)}(t)$, and summing over a frequency domain to obtain $y_1(t)$ and $y_2(t)$.

[0094] step 2.2: optimizing an objective function;

[0095] The optimization should aim to minimize the overall dynamic response of the structure. Key indicators that can reflect the overall dynamic response of the structure include parameters such as the pier bottom bending moment and the main girder displacement. The pier bottom bending moment and the main girder displacement are correlated, and optimization targets thereof are the stiffness ratio between the pile foundation and the bridge pier, as well as the mass ratio between the pile cap and the main girder.

Therefore, by optimizing the objective function with parameters such as the pier bottom bending moment and the main girder displacement, a reasonable matching of main girder mass-bridge pier stiffness-pile cap mass-pile foundation stiffness can be achieved, thereby optimizing the design of the pile foundation seismic isolation system. In the seismic design of bridges, the pier bottom bending moment is often the key parameter controlling the design. In this embodiment, the pier bottom bending moment is used as the indicator to evaluate the overall response of the structure. As shown in FIG. **6**, the pier bottom bending moment:

$$M(t)=k_\alpha L_\alpha(y_1(t)-y_2(t)) \tag{20}$$

[0096] Similarly, for any frequency component θ,

$$M^{(\theta)}(t)=k_\alpha L_\alpha(y_1^{(\theta)}(t)-y_2^{(\theta)}(t)) \tag{21}$$

[0097] substitute equations (19) and equations (20) into equation (22) to obtain $M^{(\theta)}(t)$, which is expressed in the form of a transfer function H(θ) as:

$$M(t)=\int_{-\infty}^{\infty}H(\theta)P_0e^{i\theta t}d\theta \tag{22}$$

where,

$$H(\theta)=k_\alpha L_\alpha\frac{k_\beta-m_\beta\theta^2+ic_\beta\theta}{\begin{gathered}-k_\alpha m_\alpha\theta^2+k_\alpha k_\beta-k_\alpha m_\beta\theta^2-k_\beta m_\alpha\theta^2-\\ c_\alpha c_\beta\theta^2+m_\alpha m_\beta\theta^4+\\ i(k_\beta c_\alpha\theta+k_\alpha c_\beta\theta-m_\alpha c_\alpha\theta^3-m_\beta c_\alpha\theta^3-m_\alpha c_\beta\theta^3)\end{gathered}} \tag{23}$$

[0098] Since seismic motion is a zero-mean random process, M(t) is also a zero-mean random process, when the variance σ_m^2 of M(t) is minimized, M(t) is also minimized, and according to the definition of the variance, it can be obtained that

$$\sigma_m^2=E[|M(t)|^2] \tag{24}$$

[0099] substituting equation (23) into equation (24) and performing non-dimensionalization to obtain a non-dimensional form I of the variance σ_m^2:

$$I=\frac{1}{2\pi}\int_{-\infty}^{\infty}|H(g)|^2dg \tag{25}$$

[0100] where, g represents a ratio

$$\left(\frac{\theta}{\omega_\alpha}\right)$$

of a seismic excitation to a structural frequency; and H(g) is the non-dimensional form of the transfer function H(θ):

$$H(g)=\frac{(f^2-g^2)+2i\frac{c_\beta}{c_{c\beta}}gf}{\begin{gathered}\left(-\frac{1}{\mu}g^2+(g^2-f^2)(g^2-1)+\frac{4c_\alpha c_\beta}{c_{c\alpha}c_{c\beta}}fg^2\right)+\\ i\left(2\frac{c_\alpha}{c_{c\alpha}}g\left(f^2-\frac{1}{\mu}g^2-g^2\right)+2\frac{c_\beta}{c_{c\beta}}fg(1-g^2)\right)\end{gathered}} \tag{26}$$

where,

$$\mu=\frac{m_\beta}{m_\alpha}\ \omega_\alpha^2=\frac{k_\alpha}{m_\alpha}\ \omega_\beta^2=\frac{k_\beta}{m_\beta}\ f=\frac{\omega_\beta}{\omega_\alpha}$$
$$g=\frac{\theta}{\omega_\alpha}c_{c\alpha}=2m_\alpha\omega_\beta c_{c\beta}=2m_\beta\omega_\beta\frac{k_\beta}{k_\alpha}=\mu f^2 \tag{27}$$

[0101] in equation (26), p represents a mass ratio of a pile cap to a main girder equivalent mass point, ω_α represents a natural vibration frequency of the main girder and the bridge pier, ω_β represents a natural vibration frequency of the pile cap and the pile foundation, f represents a frequency ratio between the pile foundation and the pier column, g represents a frequency ratio between an external load excitation frequency and the pier column, $c_{c\alpha}$ represents critical damping of the bridge pier, and $c_{c\beta}$ represents critical damping of the pile foundation.

[0102] For the convenience of calculation, equation (25) is organized as:

$$H(g)=\frac{B_0+igB_1-g^2B_2}{A_0+igA_1-g^2A_2-ig^3A_3+g^4A_4} \tag{28}$$

where,

$$B_0=f^2;\ B_1=2\frac{c_\beta}{c_{c\beta}}f;\ B_2=1;\ A_0=f^2;\ A_1=2\frac{c_\alpha}{c_{c\alpha}}f^2+2\frac{c_\beta}{c_{c\beta}}f$$
$$A_2=\frac{1}{\mu}+f^2+1-\frac{4c_\alpha c_\beta}{c_{c\alpha}c_{c\beta}}f;\ A_3=2\frac{1}{\mu}\frac{c_\alpha}{c_{c\alpha}}+2\frac{c_\alpha}{c_{c\alpha}}+2\frac{c_\beta}{c_{c\beta}}f;\ A_4=1 \tag{29}$$

[0103] equation (28) is substituted into equation (25) and then integrated, and let

$$\xi_1=\frac{c_1}{c_{c\alpha}}\text{ and }\xi_2=\frac{c_2}{c_{c\beta}},$$

and a formula expression for the non-dimensional form I of the variance σ^2 is obtained, which is the optimization objective function:

$$I=\frac{\left[\begin{gathered}-A_0A_1A_4B_2^2-A_0A_3A_4(B_1^2-2B_0B_2)+\\ A_4B_0^2(A_1A_4-A_2A_3)\end{gathered}\right]}{2A_0A_4(A_0A_3^2+A_1^2A_4-A_1A_2A_3)} \tag{30}$$

-continued

$$= \frac{M(\mu, f, \xi_1, \xi_2)}{M(\mu, f, \xi_1, \xi_2)}$$

[0104] Step 3: parameter optimization

[0105] For equation (30), since the damping ratios ξ_1 and ξ_2 of a concrete structure can be taken as 0.05, I is only related to the frequency ratio f and the mass ratio μ. A three-dimensional graph thereof can be plotted, as shown in FIG. **7**. From the three-dimensional graph of I(f, μ), it can be seen that for a constant mass ratio μ, the variance decreases first and then increases as the frequency ratio f increases. Therefore, for a constant mass ratio, there is a minimum value of the variance, and the corresponding frequency ratio at this minimum value is the optimal frequency ratio. By optimizing a variance equation, the optimal frequency ratio f corresponding to each mass ratio μ can be determined.

[0106] For a continuous rigid-frame bridge, since the ranges of variation for μ and f are not large and the velocity requirement is not high, the optimal frequency ratio for the pile foundation and the bridge pier can be determined by means of an enumeration method. The non-dimensional form I of the variance σ^2 has two variables μ and f. By treating f as a function of μ, the problem is transformed into a two-dimensional optimization problem. On this basis, a relational expression for the optimal frequency ratio and the mass ratio can be determined.

[0107] The optimization process using the enumeration method is as follows:

[0108] (1) determining the mass ratio μ_i;

[0109] (2) traversing the frequency ratio f;

[0110] (3) calculating the variance I(f, μ);

[0111] (4) minimizing the variance I;

[0112] (5) obtaining the corresponding $I(f_i, \mu_i)$;

[0113] (6) updating $\mu_i=\mu_i+\Delta\mu$; and simultaneously obtaining the relationship f(u) between the optimal frequency ratio and the mass ratio.

[0114] Through the optimization process mentioned above, the optimal frequency ratio f corresponding to each mass ratio μ when the variance is minimized is determined and plotted in FIG. **9**. Through curve fitting based on the above data, an explicit expression for the optimal frequency ratio f is obtained.

$$f = 1.108e^{-0.02755\mu} - 1.116e^{-0.4368\mu} \tag{31}$$

[0115] A correlation coefficient of a fitting curve R2=0.993, indicates a high correlation between the two. This represents a calculation formula for the optimal frequency ratio of the pile foundation and the bridge pier with the pier bottom bending moment as an optimization objective; after the optimal frequency ratio f is obtained, the total stiffness correction value $k_{\beta 1}$ for the 2-degree-of-freedom pile foundation integration is calculated using the following formula:

$$k_{\beta 1} = \frac{m_\beta k_\alpha f^2}{m_\alpha}. \tag{32}$$

[0116] Step 4: calculation of the pile diameter;

[0117] calculating the ratio of masses m_β and m_α according to equation (27) to obtain μ=0.66, substituting the mass ratio μ into equation (31) to obtain the optimal frequency ratio f of the pile foundation, substituting parameters into equation (32) to obtain the total stiffness correction value $k_{\beta 1}$ of the 2-degree-of-freedom pile foundation integration, calculating corrected bridge pier and pile foundation stiffness k_{41} and k_{51}, back-calculating I_{41} and I_{51} according to equation (1), and inferring the corresponding pile foundation layout and the optimal diameter based on I_{41} and I_{51}. Calculation formulas for k_{41} and k_{51} are as follows:

$$k_{41} = \frac{k_{\beta 1} k_4}{k_\beta} \tag{33}$$

$$k_{51} = \frac{k_{\beta 1} k_5}{k_\beta}$$

[0118] The above embodiments only express specific implementations of the present application, and are described in more detail, but are not to be construed as a limitation to the scope of protection of the present application. It is to be noted that several variations and modifications can also be made by those of ordinary skill in the art without departing from the concepts the technical solutions of the present application, which all fall within the scope of protection of the present application.

What is claimed is:

1. A tuned damping method for seismic isolation of pile foundations in a deep-water long-span continuous rigid frame bridge, comprising the following steps:

step 1: simplifying a deep-water long-span continuous rigid frame bridge model into a 2-degree-of-freedom dynamical model;

step 2: constructing an optimization objective function based on the 2-degree-of-freedom dynamical model;

step 3: performing parameter optimization based on the optimization objective function to obtain an optimal frequency ratio f, and based on the optimal frequency ratio f to calculate a total stiffness correction value $k_{\beta 1}$ of a 2-degree-of-freedom pile foundation integration;

step 4: calculating corrected equivalent moments of inertia I_{41} and I_{51} for a pile foundation based on the total stiffness correction value $k_{\beta 1}$ of the 2-degree-of-freedom pile foundation integration, and inferring a corresponding pile foundation layout and an optimal diameter based on the corrected equivalent moments of inertia I_{41} and I_{51}.

2. A tuned damping method for seismic isolation of pile foundations in a deep-water long-span continuous rigid frame bridge according to claim **1**, wherein the step 1 comprises:

step 1.1: simplifying the deep-water long-span continuous rigid frame bridge model into a 3-degree-of-freedom dynamical model;

step 1.2: simplifying the 3-degree-of-freedom dynamical model into the 2-degree-of-freedom dynamical model.

3. A tuned damping method for seismic isolation of pile foundations in a deep-water long-span continuous rigid frame bridge according to claim **2**, wherein the step 1.1 comprises:

letting anti-thrust stiffness of a pile group in the deep-water long-span continuous rigid frame bridge model be k_p, and solving by means of a displacement method or finite element simulation method to obtain an equivalent moment of inertia I_p:

$$I_p = \frac{k_p L_p^3}{12 E_p} \tag{1}$$

where, L_p is a pile foundation length, and E_p is the elastic modulus of pile foundation;

basing on the equivalent moment of inertia I_p to simplify the deep-water long-span continuous rigid frame bridge model into the 3-degree-of-freedom dynamical model, with parameters comprising a pier column stiffness, a pile foundation stiffness, and a mass parameters, and specific formulas for the parameters are as follows:

the pier column stiffness comprises a pier column longitudinal stiffness k_2 and a pile column transverse stiffness k_3:

$$k_2 = \frac{3E_2 I_2}{L_2^3} + \Delta k \quad k_3 = \frac{3E_3 I_3}{L_3^3} + \Delta k \tag{2}$$

the pile foundation stiffness comprises a pier foundation longitudinal stiffness k_4 and a pile foundation transverse stiffness k_5, which can be calculated using a force method or a finite element method;

main girder bending restraint correction term:

$$\Delta k = \frac{6E_2 I_2 (6E_1 I_1 I_2 + E_2 I_2 L_1)}{L_2^3 (3E_1 I_1 L_2 + 2E_2 I_2 L_1)} - \frac{3E_2 I_2}{L^3} \tag{3}$$

the mass parameters comprises a pier top mass and a pier bottom mass:

the pier top mass:

$$m_1 = \frac{m_b}{2} + \frac{m_{p2}}{2} \quad m_2 = \frac{m_b}{2} + \frac{m_{p3}}{2} \tag{4}$$

the pier bottom mass:

$$m_3 = \frac{m_c}{2} + \frac{m_{p2}}{2} + \frac{m_{p4}}{2} \qquad m_4 = \frac{m_c}{2} + \frac{m_{p3}}{2} + \frac{m_{p5}}{2} \tag{5}$$

where, I_2 and I_3 are moments of inertia of cross-sections of pier columns, I_4 and I_5 are calculated according to equation (1), m_1 is the sum of main girder mass and pier column mass on a left span of a bridge, m_2 is the sum of main girder mass and pier column mass on a right span of the bridge, m_3 is the sum of pier column mass and pile foundation mass on the left span of the bridge, m_4 is the sum of pier column mass and pile foundation mass on the right span of the bridge, m_b is total mass of a main girder, m_{p2} and m_{p3} are masses of bridge piers respectively, m_{p4} and m_{p5} are masses of pile foundations respectively, L_1 is the length of the main girder, L_2 and L_3 are the lengths of the bridge piers, L_4 and L_5 are the lengths of the pile foundations, E_1 is the elastic modulus of the main girder, E_2 and E_3 are the elastic modulus of the bridge piers, and L_4 and L_5 are the elastic modulus of the pile foundations;

a mass matrix M and a stiffness matrix K of the 3-degree-of-freedom dynamical model are:

$$M = \begin{bmatrix} m_1 + m_2 & 0 & 0 \\ 0 & m_3 & 0 \\ 0 & 0 & m_4 \end{bmatrix} \tag{6}$$

$$K = \begin{bmatrix} k_2 + k_3 + 2\Delta k & -k_2 - \Delta k & -k_3 - \Delta k \\ -k_2 - \Delta k & k_4 + k_2 + \Delta k & 0 \\ -k_3 - \Delta k & 0 & k_5 + k_3 + \Delta k \end{bmatrix}. \tag{7}$$

4. A tuned damping method for seismic isolation of pile foundations in a deep-water long-span continuous rigid frame bridge according to claim **2**, where in the step 1.2 comprises: assuming the deep-water long-span continuous rigid frame bridge model to be a symmetric structure, that is: $E_2I_2=E_3I_3$, $E_4I_4=E_5I_5$, $m_1=m_2$, $m_3=m_4$, $L_2=L_3$, and $L_4=L_5$, and further simplifying the 3-degree-of-freedom dynamical model into the 2-degree-of-freedom dynamical model, with the mass matrix M and stiffness matrix K thereof as shown in equations (8) and (9) respectively:

$$M = \begin{bmatrix} M_\alpha & 0 \\ 0 & M_\beta \end{bmatrix} \tag{8}$$

$$K = \begin{bmatrix} k_\alpha & -k_\alpha \\ -k_\alpha & k_\alpha + k_\beta \end{bmatrix} \tag{9}$$

where,

$$L_\alpha = \frac{L_2 + L_3}{2}, \quad L_\beta = \frac{L_4 + L_5}{2} \tag{10}$$

$$M_\alpha = m_1 + m_2, \quad M_\beta = m_3 + m_4 \tag{11}$$

$$k_\alpha = 4(k_2 + k_3) \tag{12}$$

$$k_\beta = k_4 + k_5 \tag{13}$$

Where, k_α is total stiffness of a 2-degree-of-freedom pier column integration, and k_β is total stiffness of the 2-degree-of-freedom pile foundation integration.

5. A tuned damping method for seismic isolation of pile foundations in a deep-water long-span continuous rigid frame bridge according to claim **1**, wherein the step 2 comprises:

step 2.1: constructing motion equations for a 2-degree-of-freedom system based on the 2-degree-of-freedom dynamical model;

step 2.2: constructing an optimization objective function based on the motion equations for the 2-degree-of-freedom system.

6. A tuned damping method for seismic isolation of pile foundations in a deep-water long-span continuous rigid frame bridge according to claim **5**, wherein the step 2.1 comprises: considering damping of the bridge piers and the pile foundations, and enabling the bridge structure to undergo random vibration under the action of P(t), wherein y_1 and y_2 represent 2-degree-of-freedom displacement time history functions relative to a foundation, and motion equations for the 2-degree-of-freedom dynamical model can be obtained by means of a direct equilibrium method as follows:

$$m_\alpha \ddot{y}_\alpha + c_\alpha(\dot{y}_\alpha - \dot{y}_\beta) + k_\alpha(y_\alpha - y_\beta) = P(t) \tag{14}$$
$$m_\beta \ddot{y}_\beta + c_\beta \dot{y}_\beta + c_\alpha(\dot{y}_\beta - \dot{y}_\alpha) + k_\beta y_\beta + k_\alpha(y_\beta - y_\alpha) = 0$$

where c_i is a damping coefficient; y_i is the relative displacement of a mass point; $\dot{y}_i$ is the relative velocity of the mass point; $\ddot{y}_i$ is a relative acceleration of the mass point; P(t) is an inertial force caused by ground motion, and P(t) is a random excitation which can be decomposed into the superposition of a series of harmonic components as follows:

$$P(t) = \int_{-\infty}^{\infty} p_0 e^{i\theta t} d\theta \tag{15}$$

for any frequency component θ, $P^{(\Theta)}(t)=p_0e^{i\Theta t}$ is a harmonic excitation, and resulting displacements $y_1^{(\Theta)}(t)$ and $y_2^{(\Theta)}$ (t) are also harmonic variables and can be expressed as follows:

$$y_1^{(\theta)}(t) = Y_1 e^{i\theta t},\ y_2^{(\theta)}(t) = Y_2 e^{i\theta t} \tag{16}$$

substituting $P^{(\Theta)}(t)$ and equation (16) into equation (14) to obtain

$$-m_\alpha \theta^2 Y_1 + c_\alpha i\theta(Y_1 - Y_2) + k_\alpha(Y_1 - Y_2) = p_0 \tag{17}$$
$$-m_\beta \theta^2 Y_2 + c_\beta i\theta Y_2 + c_\alpha i\theta(Y_2 - Y_1) + k_\beta Y_\beta + k_\alpha(Y_2 - Y_1) = 0$$

solving to obtain amplitudes of $y_1(t)$ and $y_2(t)$:

$$Y_1 = \frac{-p_0(k_\alpha + k_\beta - m_\beta\theta^2 + c_\alpha\theta i + c_\beta\theta i)}{\begin{gathered}k_\alpha m_\alpha\theta^2 - k_\alpha k_\beta + k_\alpha m_\beta\theta^2 + k_\beta m_\alpha\theta^2 + c_\alpha c_\beta\theta^2 - m_\alpha m_\beta\theta^4 - \\ k_\beta c_\alpha\theta i - k_\alpha c_\beta\theta i + m_\alpha c_\alpha\theta^3 i + m_\beta c_\alpha\theta^3 i + m_\alpha c_\beta\theta^3 i\end{gathered}} \tag{18}$$

$$Y_2 = \frac{-p_0(k_\alpha + c_\alpha\theta i)}{\begin{gathered}k_\alpha m_\alpha\theta^2 - k_\alpha k_\beta + k_\alpha m_\beta\theta^2 + k_\beta m_\alpha\theta^2 + c_\alpha c_\beta\theta^2 - m_\alpha m_\beta\theta^4 - \\ k_\beta c_\alpha\theta i - k_\alpha c_\beta\theta i + m_\alpha c_\alpha\theta^3 i + m_\beta c_\alpha\theta^3 i + m_\alpha c_\beta\theta^3 i\end{gathered}} \tag{19}$$

substituting into equations (16) to obtain $y_1^{(\Theta)}(t)$ and $y_2^{(\Theta)}(t)$, and summing $y_1^{(\Theta)}(t)$ and $y_2^{(\Theta)}(t)$ over a frequency domain to obtain $y_1(t)$ and $y_2(t)$.

7. A tuned damping method for seismic isolation of pile foundations in a deep-water long-span continuous rigid frame bridge according to claim **6**, where in the step 2.2 comprises: letting a pier bottom bending moment as an indicator to evaluate an overall structural response:

$$M(t) = k_\alpha L_\alpha(y_1(t) - y_2(t)) \tag{20}$$

for any frequency component θ,

$$M^{(\theta)}(t) = k_\alpha L_\alpha\left(y_1^{(\theta)}(t) - y_2^{(\theta)}(t)\right) \tag{21}$$

substituting equations (18) and (19) into equation (21) to obtain $M^{(\Theta)}(t)$, and expressing $M^{(\Theta)}(t)$ in the form of a transfer function H(θ) as follows:

$$M(t) = \int_{-\infty}^{\infty} H(\theta) P_0 e^{i\theta t} d\theta \tag{22}$$

where,

$$H(\theta) = k_\alpha L_\alpha \frac{k_\beta - m_\beta\theta^2 + ic_\beta\theta}{\begin{gathered}-k_\alpha m_\alpha\theta^2 + k_\alpha k_\beta - k_\alpha m_\beta\theta^2 - k_\beta m_\alpha\theta^2 - c_\alpha c_\beta\theta^2 + m_\alpha m_\beta\theta^4 + \\ i\left(k_\beta c_\alpha\theta + k_\alpha c_\beta\theta - m_\alpha c_\alpha\theta^3 - m_\beta c_\alpha\theta^3 - m_\alpha c_\beta\theta^3\right)\end{gathered}} \tag{23}$$

when a variance σ_m^2 of M(t) is minimized, M(t) is also minimized, and let the variance σ_m^2 be:

$$\sigma_m^2 = E\left[|M(t)|^2\right] \tag{24}$$

substituting equation (23) into equation (24) and performing non-dimensionalization to obtain a non-dimensional form I of the variance σ_m^2:

$$I = \frac{1}{2\pi}\int_{-\infty}^{\infty} |H(g)|^2 dg \tag{25}$$

where, g represents a ratio

$$\left(\frac{\theta}{\omega_\alpha}\right)$$

of a seismic excitation to a structural frequency; and H(g) is the non-dimensional form of the transfer function H(θ):

$$H(g) = \frac{(f^2 - g^2) + 2i\frac{c_\beta}{c_{c\beta}}gf}{\begin{gathered}\left(-\frac{1}{\mu}g^2 + (g^2 - f^2)(g^2 - 1) + \frac{4c_\alpha c_\beta}{c_{c\alpha}c_{c\beta}}fg^2\right) + \\ i\left(2\frac{c_\alpha}{c_{c\alpha}}g\left(f^2 - \frac{1}{\mu}g^2 - g^2\right) + 2\frac{c_\beta}{c_{c\beta}}fg(1 - g^2)\right)\end{gathered}} \tag{26}$$

where,

$$\mu = \frac{m_\beta}{m_\alpha}\omega_\alpha^2 = \frac{k_\alpha}{m_\alpha}\omega_\beta^2 = \frac{k_\beta}{m_\beta}f = \frac{\omega_\beta}{\omega_\alpha} \tag{27}$$

-continued

$$g=\frac{\theta}{\omega_\alpha}c_{c\alpha}=2m_\alpha\omega_\beta c_{c\beta}=2m_\beta\omega_\beta\frac{k_\beta}{k_\alpha}=\mu f^2$$

in equation (26), μ represents a mass ratio of a pile cap to a main girder equivalent mass point, ω_α represents a natural vibration frequency of the main girder and the bridge pier, ω_β represents a natural vibration frequency of the pile cap and the pile foundation, f represents a frequency ratio between the pile foundation and the pier column, g represents a frequency ratio between an external load excitation and the pier column, $c_{c\alpha}$ represents critical damping of the bridge pier, and $c_{c\beta}$ represents critical damping of the pile foundation;

organizing equation (25) as:

$$H(g)=\frac{B_0+igB_1-g^2B_2}{A_0+igA_1-g^2A_2-ig^3A_3+g^4A_4} \tag{28}$$

where,

$$\begin{aligned}B_0&=f^2;\\B_1&=2\frac{c_\beta}{c_{c\beta}}f;\\B_2&=1;\\A_0&=f^2;\\A_1&=2\frac{c_\alpha}{c_{c\alpha}}f^2+2\frac{c_\beta}{c_{c\beta}}f\\A_2&=\frac{1}{\mu}+f^2+1-\frac{4c_\alpha c_\beta}{c_{c\alpha}c_{c\beta}}f;\\A_3&=2\frac{1}{\mu}\frac{c_\alpha}{c_{c\alpha}}+2\frac{c_\alpha}{c_{c\alpha}}+2\frac{c\beta}{c_{c\beta}}f;\\A_4&=1\end{aligned} \tag{29}$$

substituting equation (28) into equation (25) for integration, letting

$$\xi_1=\frac{c_1}{c_{c\alpha}}\text{ and }\xi_2=\frac{c_2}{c_{c\beta}},$$

and obtaining the non-dimensional form I of the variance σ^2, which serves as the optimization objective function:

$$I=\frac{\left[-A_0A_1A_4B_2^2-A_0A_3A_4\left(B_1^2-2B_0B_2\right)+A_4B_0^2(A_1A_4-A_2A_3)\right]}{2A_0A_4\left(A_0A_3^2+A_1^2A_4-A_1A_2A_3\right)}=\frac{M(\mu,f,\xi_1,\xi_2)}{N(\mu,f,\xi_1,\xi_2)}. \tag{30}$$

8. A tuned damping method for seismic isolation of pile foundations in a deep-water long-span continuous rigid frame bridge according to claim **1**, where in the step 3 comprises: optimizing the variance by using an enumeration method, finding an optimal frequency ratio f corresponding to each mass ratio μ when the variance is minimized, and performing curve fitting to obtain an explicit expression for the optimal frequency ratio f:

$$f=1.108e^{-0.02755\mu}-1.116e^{-0.4368\mu} \tag{31}$$

based on the optimal frequency ratio f, calculating a total stiffness correction value $k_{\beta1}$ of a 2-degree-of-freedom pile foundation integration:

$$k_{\beta1}=\frac{m_\beta k_\alpha f^2}{m_\alpha}. \tag{32}$$

9. A tuned damping method for seismic isolation of pile foundations in a deep-water long-span continuous rigid frame bridge according to claim **8**, where in the optimization process using the enumeration method comprises the following steps:

(1) determining the mass ratio μ;
(2) traversing the frequency ratio f;
(3) calculating the variance I(f, μ);
(4) minimizing the variance I;
(5) obtaining the corresponding $I(f_i, \mu_i)$; and
(6) updating $\mu_i=\mu_i+\Delta\mu$; and simultaneously obtaining the relationship f(u) between the optimal frequency ratio and the mass ratio.

10. A tuned damping method for seismic isolation of pile foundations in a deep-water long-span continuous rigid frame bridge according to claim **1**, where in the step 4 comprises:

calculating the mass ratio μ according to equation (27), substituting the mass ratio μ into equation (31) to obtain an optimal frequency ratio f of the pile foundation, according to equation (32) to obtain the total stiffness correction value $k_{\beta1}$ for the 2-degree-of-freedom pile foundation integration, calculating corrected bridge pier and pile foundation stiffness k_{41} and k_{51}, back-calculating the equivalent moments of inertia I_{41} and I_{51} of the pile foundation according to equation (1), and based on the equivalent moments of inertia I_{41} and I_{51}, inferring the corresponding pile foundation layout and the optimal diameter, where calculation formulas for k_{41} and k_{51} are as follows:

$$\begin{aligned}k_{41}&=\frac{k_{\beta1}k_4}{k_\beta}\\k_{51}&=\frac{k_{\beta1}k_5}{k_\beta}\end{aligned}. \tag{33}$$

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